

Project Name: New project

CONTENTS

Bill Of Material	5
Controller	5
Cartridge	5
Hardware Configuration	6
MyController - TM221CE24R	6
Digital Inputs	6
Digital Outputs	6
Analog Inputs	6
Fast Counters	7
High Speed Counters	7
ETH1	7
SL1 (Serial line)	8
Cartridges	9
TMC2AI2	9
Analog Inputs	9
Software Configuration	10
Constant Words	10
KW	10
KD	10
KF	10
Network Objects	11
Input Assembly (Ethernet/Ip)	11
Output Assembly (Ethernet/Ip)	11
Input Registers (Modbus Tcp)	11
Output Registers (Modbus Tcp)	11
Digital inputs (IOScanner)	11
Digital outputs (IOScanner)	11
Input registers (IOScanner)	11
Output registers (IOScanner)	11
Software Objects	12
Timers	12
Counters	12
LIFO/FIFO Registers	12
Drums	12
Shift Bit Registers	12
Step Counters	12
Schedule Blocks	12
RTC	12

Project Name: New project

07/14/2026 - 09:54 PM

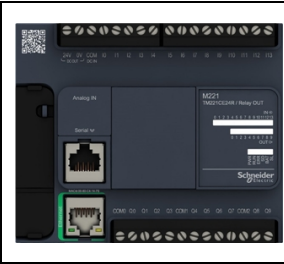
2/30

PID	12
Grafcet Steps	13
Program	14
Behavior	14
Memory Consumption	14
Application Architecture	14
Master Task	14
Periodic Task	14
POU	15
Master Task	15
1 - POU_Inputs	15
Rung0	15
Rung1	15
Rung2	15
Rung3	15
Rung4	15
Rung5	16
Rung6	16
Rung7	16
Rung8	16
Rung9	16
2 - POU_Scaling	17
Rung0	17
3 - Main_Program	18
Rung0	18
Rung1	18
Rung2	18
Rung3	18
4 - POU_Modes	19
Rung0	19
Rung1	19
Rung2	19
Rung3	19
5 - POU_Valve_Logic	20
Rung0	20
Rung1	20
Rung2	20
Rung3	21
Rung4	21

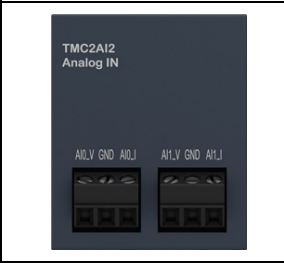
Rung5	21
6 - POU_Outputs	22
Rung0	22
Rung1	22
Rung2	22
Rung3	22
Rung4	22
Rung5	23
Rung6	23
Rung7	23
Rung8	23
Rung9	24
Symbols	25
Cross-Reference Table	27

BILL OF MATERIAL

Controller

	Reference	TM221CE24R
	Description	TM221CE24R (screw) 14 digital inputs, 10 relay outputs (2 A), 2 analog inputs, 1 serial line port, 1 Ethernet port, 100-240 Vac power supply controller with removable terminal blocks.
	Power supplied to the IO bus	5V: 520 mA / 24V: 160 mA

Cartridge

	Reference	TMC2AI2
	Description	TMC2AI2 TMC2 cartridge with 2 analog voltage or current inputs (0 - 10V, 0-20mA, 4 - 20mA), 12 bits.

HARDWARE CONFIGURATION

MyController - TM221CE24R

Digital Inputs

Used	Address	Filtering	Latch	Run/Stop	Events	Priority	Subroutine
X	%I0.0	3 ms			Not Used		
X	%I0.1	3 ms			Not Used		
X	%I0.2	3 ms			Not Used		
X	%I0.3	3 ms			Not Used		
X	%I0.4	3 ms			Not Used		
X	%I0.5	3 ms			Not Used		
X	%I0.6	3 ms			Not Used		
X	%I0.7	3 ms			Not Used		
	%I0.8	3 ms			Not Used		
	%I0.9	3 ms			Not Used		
	%I0.10	3 ms			Not Used		
	%I0.11	3 ms			Not Used		
	%I0.12	3 ms			Not Used		
	%I0.13	3 ms			Not Used		

Digital Outputs

Used	Address	Status Alarm	Fallback value	Used by
X	%Q0.0		0	User logic
X	%Q0.1		0	User logic
X	%Q0.2		0	User logic
X	%Q0.3		0	User logic
X	%Q0.4		0	User logic
X	%Q0.5		0	User logic
X	%Q0.6		0	User logic
X	%Q0.7		0	User logic
X	%Q0.8		0	User logic
X	%Q0.9		0	User logic

Analog Inputs

Used	Address	Type	Scope	Range	Filter	Sampling
	%IW0.0	0 - 10 V	Normal	0-1000	0	
	%IW0.1	0 - 10 V	Normal	0-1000	0	

Fast Counters

Used	Address	Input	Configured	Preset	Double Word
	%FC0	%I0.2	NotUsed	0	
	%FC1	%I0.3	NotUsed	0	
	%FC2	%I0.4	NotUsed	0	
	%FC3	%I0.5	NotUsed	0	

High Speed Counters

Used	Address	Type
	%HSC0	Not Configured
	%HSC1	Not Configured
	%HSC2	Not Configured
	%HSC3	Not Configured

ETH1

Device name:	M221
IP Mode:	Fixed
IP address:	0.0.0.0
Subnet mask:	0.0.0.0
Gateway address:	0.0.0.0
Transfer Rate:	Auto
Security Parameters:	Programming protocol disabled
	Auto discovery protocol disabled
	Modbus server disabled
	EtherNet/IP protocol disabled

SL1 (Serial line)

Physical Settings

Device:	None
Baud rate:	19200
Parity:	Even
Data bits:	8
Stop bits:	1
Physical medium:	RS-485
Polarization:	No

Protocol Settings

Protocol:	Modbus
Response timeout (× 100 ms):	10
Time between frames (ms):	10
Transmission mode:	RTU
Addressing:	Slave
Address:	1

Cartridges

TMC2AI2

Analog Inputs

Used	Address	Type	Scope	Range	Filter	Sampling
X	%IW0.100	4 - 20 mA	Normal	4000- 20000	0	
	%IW0.101	4 - 20 mA	Normal	4000- 20000	0	

SOFTWARE CONFIGURATION

Constant Words

KW

Allocation: Automatic

Allocated: 0

Used	Equ Used	Address	Symbol	Value	Comment
------	----------	---------	--------	-------	---------

KD

Allocation: Automatic

Allocated: 0

Used	Equ Used	Address	Symbol	Value	Comment
------	----------	---------	--------	-------	---------

KF

Allocation: Automatic

Allocated: 0

Used	Equ Used	Address	Symbol	Value	Comment
------	----------	---------	--------	-------	---------

Network Objects

Input Assembly (Ethernet/Ip)

Used	Address	Symbol	Fallback value	Comment
------	---------	--------	----------------	---------

Output Assembly (Ethernet/Ip)

Used	Address	Symbol	Comment
------	---------	--------	---------

Input Registers (Modbus Tcp)

Used	Address	Symbol	Fallback value	Comment
------	---------	--------	----------------	---------

Output Registers (Modbus Tcp)

Used	Address	Symbol	Comment
------	---------	--------	---------

Digital inputs (IOScanner)

Used	Address	Channel	Symbol	Comment
------	---------	---------	--------	---------

Digital outputs (IOScanner)

Used	Address	Channel	Fallback value	Symbol	Comment
------	---------	---------	----------------	--------	---------

Input registers (IOScanner)

Used	Address	Channel	Symbol	Comment
------	---------	---------	--------	---------

Output registers (IOScanner)

Used	Address	Channel	Fallback value	Symbol	Comment
------	---------	---------	----------------	--------	---------

Software Objects

Timers

Allocation: Automatic

Allocated: 2

Used	Address	Symbol	Type	Retentive	Dynamic Preset	Time Base	Preset	Comment
X	%TM0		TON			1 s	180	
X	%TM1		TON			1 s	200	DELAY_STAIRT

Counters

Allocation: Automatic

Allocated: 0

LIFO/FIFO Registers

Allocation: Automatic

Allocated: 0

Drums

Allocation: Automatic

Allocated: 0

Shift Bit Registers

Allocation: Automatic

Allocated: 0

Step Counters

Allocation: Automatic

Allocated: 0

Schedule Blocks

Allocation: Automatic

Allocated: 0

RTC

PID

Used	PID	Symbol	Type	Comment
Project Name: New project				
07/14/2026 - 09:54 PM				
12/30				

Grafcet Steps

Allocation: Automatic

Allocated: 0

PROGRAM

Behavior

Functional level:	Level 14.0
Starting mode:	Start In Run
Watchdog:	250 ms
Fallback behavior:	Fallback value
String end character:	CR (Carriage Return)

Memory consumption

A successful compilation is required to obtain memory information.

Application Architecture

Master Task

Scan mode:	Normal
POU list:	1 - POU_Inputs
	2 - POU_Scaling
	3 - Main_Program
	4 - POU_Modes
	5 - POU_Valve_Logic
	6 - POU_Outputs

Periodic Task

Period:	255 ms
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POU

Master Task

1 - POU_Inputs

Master Task

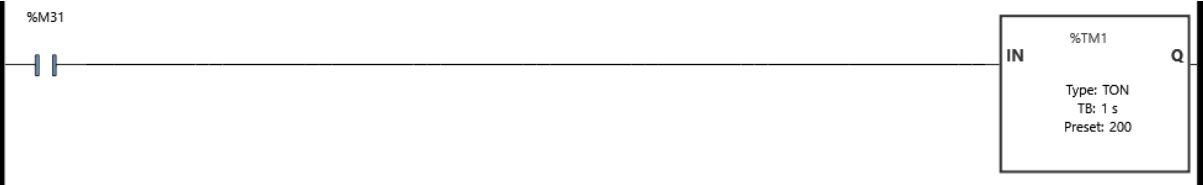
Rung0



Variables used:

%M31	POWER_DELAY_ACTIVE	
%S1	SB_WARMSTART	Indicates there was a warm start with data backup

Rung1



Variables used:

%M31	POWER_DELAY_ACTIVE	
%TM1	DELAY_START	

Rung2



Variables used:

%I0.3	DAIKIN_ALARM	
%M4	ALARM_INT	

Rung3



Variables used:

%I0.2	CHILLER_PUMP_FB	pump feedback
%M21	CHILLER_PUMP_FB_INT	

Rung4



Variables used:

%I0.4	DAIKIN_RUN_FB	Fault Signal (80-81)
%M30	DAIKIN_RUN_FB_INT	

Rung5



Legend:

1 %MW2 := %IW0.100

Variables used:

%IW0.100
%MW2 TEMP_SENSOR_RAW

Rung6



Variables used:

%I0.5 DAIKIN_MODE Operation Feedback(82-83)
%M5 DAIKIN_MODE_FB

Rung7



Variables used:

%I0.6 CIRC_PUMP_FB Water Flow Signal(70-71)
%M27 PUMP_FB_INT

Rung8



Variables used:

%I0.7 DI_VALVE_FB
%M17 VALVE_AT_CHILLER

Rung9



Variables used:

%I0.7 DI_VALVE_FB
%M18 VALVE_AT_BOILER

2 - POU_Scaling

Master Task

Rung0



Legend:

1 MF0 := ((INT_TO_REAL(%MW2) - 4000.0) / 16000.0) * 80.0 - 20.0

Variables used:

%MW2 TEMP_SENSOR_RAW

3 - Main_Program

Master Task

Rung0



Variables used:

%I0.0	MAIN_START	System Start Button
%I0.1	MAIN_STOP	Emergency Stop Button
%M0	SYSTEM_READY	
%M4	ALARM_INT	
%M31	POWER_DELAY_ACTIVE	
%M32	TIMER_DONE_MEMORY	

Rung1



Legend:

1 %TM1.Q

Variables used:

%M32	TIMER_DONE_MEMORY	
%TM1.Q	DELAY_START	

Rung2



Variables used:

%M31	POWER_DELAY_ACTIVE	
%M32	TIMER_DONE_MEMORY	

Rung3



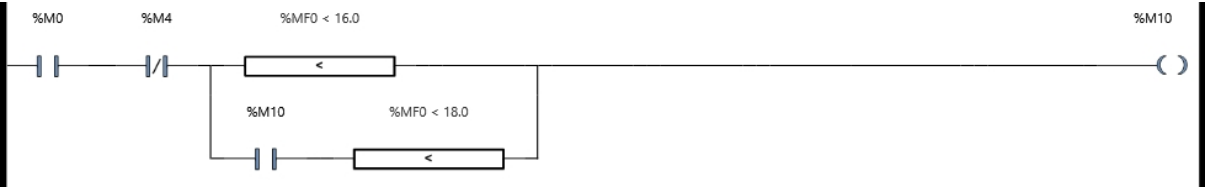
Variables used:

%M0	SYSTEM_READY	
%M4	ALARM_INT	
%M28	CIRC_PUMP_INT_CMD	

4 - POU_Modes

Master Task

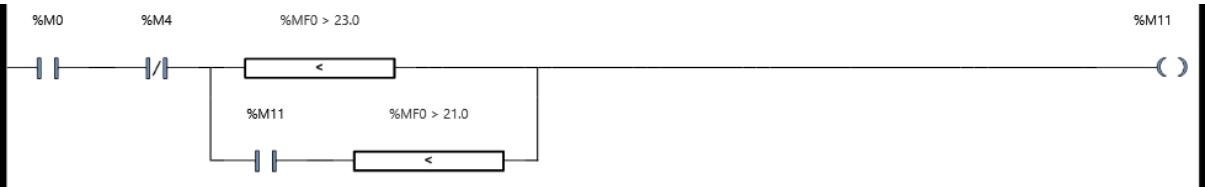
Rung0



Variables used:

%M0	SYSTEM_READY	
%M4	ALARM_INT	
%M10	HEATING_ACTIVE	
%MF0	ACTUAL_TEMP	Processed Outdoor Temp (REAL)

Rung1



Variables used:

%M0	SYSTEM_READY	
%M4	ALARM_INT	
%M11	COOLING_ACTIVE	
%MF0	ACTUAL_TEMP	Processed Outdoor Temp (REAL)

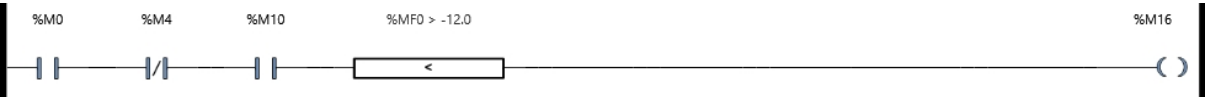
Rung2



Variables used:

%M0	SYSTEM_READY	
%M4	ALARM_INT	
%M10	HEATING_ACTIVE	
%M15	BOILER_MODE	
%MF0	ACTUAL_TEMP	Processed Outdoor Temp (REAL)

Rung3



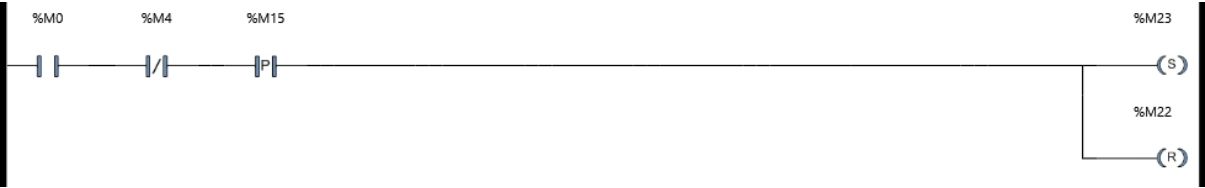
Variables used:

%M0	SYSTEM_READY	
%M4	ALARM_INT	
%M10	HEATING_ACTIVE	
%M16	CHILLER_MODE	
%MF0	ACTUAL_TEMP	Processed Outdoor Temp (REAL)

5 - POU_Valve_Logic

Master Task

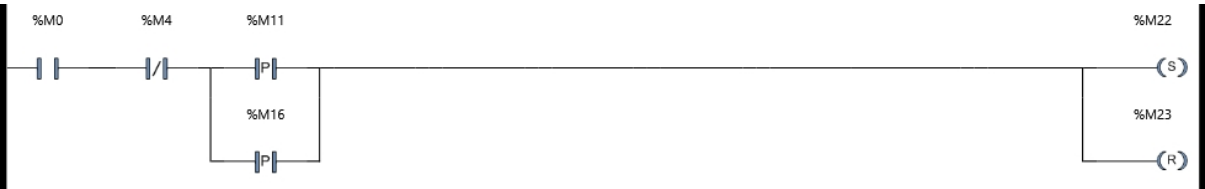
Rung0



Variables used:

%M0	SYSTEM_READY
%M4	ALARM_INT
%M15	BOILER_MODE
%M22	VALVE_TO_CHILLER
%M23	VALVE_TO_BOILER

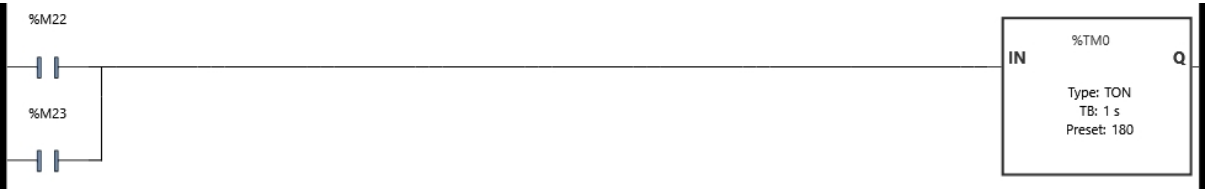
Rung1



Variables used:

%M0	SYSTEM_READY
%M4	ALARM_INT
%M11	COOLING_ACTIVE
%M16	CHILLER_MODE
%M22	VALVE_TO_CHILLER
%M23	VALVE_TO_BOILER

Rung2



Variables used:

%M22	VALVE_TO_CHILLER
%M23	VALVE_TO_BOILER
%TM0	

Rung3



Legend:

1 %TM0.Q

Variables used:

%M22 VALVE_TO_CHILLER

%M23 VALVE_TO_BOILER

%TM0.Q

Rung4



Variables used:

%M0 SYSTEM_READY

%M4 ALARM_INT

%M22 VALVE_TO_CHILLER

%M23 VALVE_TO_BOILER

Rung5



Variables used:

%M22 VALVE_TO_CHILLER

%M23 VALVE_TO_BOILER

%M24 VALVE_TIMER_ACTIVE

6 - POU_Outputs

Master Task

Rung0



Variables used:

%M28	CIRC_PUMP_INT_CMD	
%Q0.0	PUMP_START	Pump Control

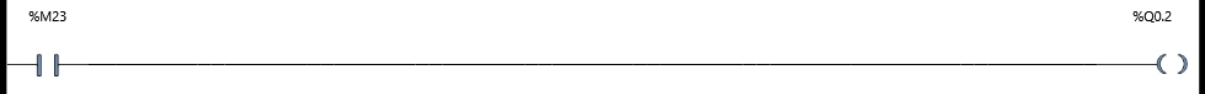
Rung1



Variables used:

%M0	SYSTEM_READY	
%M4	ALARM_INT	
%M10	HEATING_ACTIVE	
%M15	BOILER_MODE	
%M18	VALVE_AT_BOILER	
%Q0.1	BOILER_START	Daikin Remote ON/OFF(76-77)

Rung2



Variables used:

%M23	VALVE_TO_BOILER	
%Q0.2	VALVE_BOILER	Cool/Heat Selection(72-73)

Rung3



Variables used:

%M22	VALVE_TO_CHILLER	
%Q0.3	VALVE_CHILLER	Boiler Start

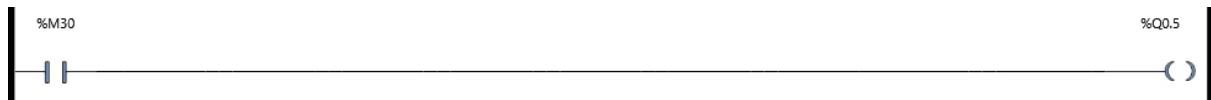
Rung4



Variables used:

%M0	SYSTEM_READY	
%Q0.4	LAMP_START	Move to Chiller Position

Rung5



Variables used:

%M30	DAIKIN_RUN_FB_INT	
%Q0.5	DAIKIN_RUN	Move to Boiler Position

Rung6



Variables used:

%M4	ALARM_INT	
%Q0.6	LAMP_ALARM	System Alarm Indicator

Rung7



Variables used:

%M0	SYSTEM_READY	
%M4	ALARM_INT	
%M5	DAIKIN_MODE_FB	
%Q0.7	LAMP_RUN	System Operation Indicator

Rung8



Variables used:

%M0	SYSTEM_READY	
%M4	ALARM_INT	
%M11	COOLING_ACTIVE	
%M16	CHILLER_MODE	
%M17	VALVE_AT_CHILLER	
%Q0.8	START_CHILLER	Heating Mode Active Indicator

Rung9



Variables used:

%M10	HEATING_ACTIVE	
%Q0.9	COOLING_HEATING	Cooling Mode Active Indicator

SYMBOLS

Used	Address	Symbol	Comment
X	%I0.0	MAIN_START	System Start Button
X	%I0.1	MAIN_STOP	Emergency Stop Button
X	%I0.2	CHILLER_PUMP_FB	pump feedback
X	%I0.3	DAIKIN_ALARM	
X	%I0.4	DAIKIN_RUN_FB	Fault Signal (80-81)
X	%I0.5	DAIKIN_MODE	Operation Feedback(82-83)
X	%I0.6	CIRC_PUMP_FB	Water Flow Signal(70-71)
X	%I0.7	DI_VALVE_FB	
X	%M0	SYSTEM_READY	
X	%M4	ALARM_INT	
X	%M5	DAIKIN_MODE_FB	
X	%M10	HEATING_ACTIVE	
X	%M11	COOLING_ACTIVE	
X	%M15	BOILER_MODE	
X	%M16	CHILLER_MODE	
X	%M17	VALVE_AT_CHILLER	
X	%M18	VALVE_AT_BOILER	
X	%M21	CHILLER_PUMP_FB_INT	
X	%M22	VALVE_TO_CHILLER	
X	%M23	VALVE_TO_BOILER	
X	%M24	VALVE_TIMER_ACTIVE	
X	%M27	PUMP_FB_INT	
X	%M28	CIRC_PUMP_INT_CMD	
X	%M30	DAIKIN_RUN_FB_INT	
X	%M31	POWER_DELAY_ACTIVE	
X	%M32	TIMER_DONE_MEMORY	
X	%MF0	ACTUAL_TEMP	Processed Outdoor Temp (REAL)
X	%MW2	TEMP_SENSOR_RAW	
X	%Q0.0	PUMP_START	Pump Control
X	%Q0.1	BOILER_START	Daikin Remote ON/OFF(76-77)
X	%Q0.2	VALVE_BOILER	Cool/Heat Selection(72-73)

Used	Address	Symbol	Comment
X	%Q0.3	VALVE_CHILLER	Boiler Start
X	%Q0.4	LAMP_START	Move to Chiller Position
X	%Q0.5	DAIKIN_RUN	Move to Boiler Position
X	%Q0.6	LAMP_ALARM	System Alarm Indicator
X	%Q0.7	LAMP_RUN	System Operation Indicator
X	%Q0.8	START_CHILLER	Heating Mode Active Indicator
X	%Q0.9	COOLING_HEATING	Cooling Mode Active Indicator
X	%S1	SB_WARMSTART	Indicates there was a warm start with data backup

CROSS-REFERENCE TABLE

Address	Object	Rung	Code
%I0.0....	3 - Main_Program	Rung0	-- P --
%I0.1....	3 - Main_Program	Rung0	-- --
%I0.2....	1 - POU_Inputs	Rung3	-- --
%I0.3....	1 - POU_Inputs	Rung2	-- --
%I0.4....	1 - POU_Inputs	Rung4	-- --
%I0.5....	1 - POU_Inputs	Rung6	-- --
%I0.6....	1 - POU_Inputs	Rung7	-- --
%I0.7....	1 - POU_Inputs	Rung8	-- --
		Rung9	-- / --
%IW0.100.	1 - POU_Inputs	Rung5	--[...]-- %MW2 := %IW0.100
%M0.....	3 - Main_Program	Rung0	--()--
			-- --
		Rung3	-- --
	4 - POU_Modes	Rung0	-- --
		Rung1	-- --
		Rung2	-- --
		Rung3	-- --
	5 - POU_Valve_Logic	Rung0	-- --
		Rung1	-- --
		Rung4	-- / --
	6 - POU_Outputs	Rung1	-- --
		Rung4	-- --
		Rung7	-- --
		Rung8	-- --
%M4.....	1 - POU_Inputs	Rung2	--()--
	3 - Main_Program	Rung0	-- / --
		Rung3	-- / --
	4 - POU_Modes	Rung0	-- / --
		Rung1	-- / --
		Rung2	-- / --
		Rung3	-- / --

Address	Object	Rung	Code		
	5 - POU_Valve_Logic	Rung0	-- / --		
		Rung1	-- / --		
		Rung4	-- --		
	6 - POU_Outputs	Rung1	-- / --		
		Rung6	-- --		
		Rung7	-- / --		
		Rung8	-- / --		
	%M5.....	1 - POU_Inputs	Rung6	--()--	
	6 - POU_Outputs	Rung7	-- --		
	%M10.....	4 - POU_Modes	Rung0	--()--	
			-- --		
		Rung2	-- --		
		Rung3	-- --		
	6 - POU_Outputs	Rung1	-- --		
		Rung9	-- --		
	%M11.....	4 - POU_Modes	Rung1	--()--	
				-- --	
			5 - POU_Valve_Logic	Rung1	-- P --
			6 - POU_Outputs	Rung8	-- --
		%M15.....	4 - POU_Modes	Rung2	--()--
5 - POU_Valve_Logic		Rung0	-- P --		
6 - POU_Outputs		Rung1	-- --		
%M16.....		4 - POU_Modes	Rung3	--()--	
5 - POU_Valve_Logic		Rung1	-- P --		
6 - POU_Outputs		Rung8	-- --		
%M17.....		1 - POU_Inputs	Rung8	--()--	
6 - POU_Outputs	Rung8	-- --			
%M18.....	1 - POU_Inputs	Rung9	--()--		
6 - POU_Outputs	Rung1	-- --			
%M21.....	1 - POU_Inputs	Rung3	--()--		
%M22.....	5 - POU_Valve_Logic	Rung0	--(R)--		
	Rung1	--(S)--			
	Rung2	-- --			
	Rung3	--(R)--			

Address	Object	Rung	Code
%M23.....	6 - POU_Outputs 5 - POU_Valve_Logic	Rung4	-- (R) --
		Rung5	-- --
		Rung3	-- --
		Rung0	-- (S) --
		Rung1	-- (R) --
		Rung2	-- --
		Rung3	-- (R) --
		Rung4	-- (R) --
		Rung5	-- --
		Rung2	-- --
%M24.....	5 - POU_Valve_Logic	Rung5	-- () --
%M27.....	1 - POU_Inputs	Rung7	-- () --
%M28.....	3 - Main_Program	Rung3	-- () --
%M30.....	6 - POU_Outputs	Rung0	-- --
	1 - POU_Inputs	Rung4	-- () --
	6 - POU_Outputs	Rung5	-- --
%M31.....	1 - POU_Inputs	Rung0	-- (S) --
		Rung1	-- --
	3 - Main_Program	Rung0	-- / --
		Rung2	-- (R) --
%M32.....	3 - Main_Program	Rung0	-- --
		Rung1	-- () --
		Rung2	-- P --
%MF0.....	2 - POU_Scaling	Rung0	--[...]-- %MF0 := ((INT_TO_REAL (%MW2) - 4000.0) / 16000.0) * 80.0 - 20.0
	4 - POU_Modes	Rung0	--[<]-- %MF0 < 16.0 --[<]-- %MF0 < 18.0
		Rung1	--[<]-- %MF0 > 23.0 --[<]-- %MF0 > 21.0
		Rung2	--[<]-- %MF0 < -15.0
		Rung3	--[<]-- %MF0 > -12.0
%MW2.....	1 - POU_Inputs	Rung5	--[...]-- %MW2 := %IW0.100
	2 - POU_Scaling	Rung0	--[...]-- %MF0 := ((INT_TO_REAL (%MW2) - 4000.0) / 16000.0) * 80.0 - 20.0

Address	Object	Rung	Code
%Q0.0....	6 - POU_Outputs	Rung0	-- () --
%Q0.1....	6 - POU_Outputs	Rung1	-- () --
%Q0.2....	6 - POU_Outputs	Rung2	-- () --
%Q0.3....	6 - POU_Outputs	Rung3	-- () --
%Q0.4....	6 - POU_Outputs	Rung4	-- () --
%Q0.5....	6 - POU_Outputs	Rung5	-- () --
%Q0.6....	6 - POU_Outputs	Rung6	-- () --
%Q0.7....	6 - POU_Outputs	Rung7	-- () --
%Q0.8....	6 - POU_Outputs	Rung8	-- () --
%Q0.9....	6 - POU_Outputs	Rung9	-- () --
%S1.....	1 - POU_Inputs	Rung0	-- --
%TM0.....	5 - POU_Valve_Logic	Rung2	%TM0
%TM0.Q...	5 - POU_Valve_Logic	Rung3	-- --
%TM1.....	1 - POU_Inputs	Rung1	%TM1
%TM1.Q...	3 - Main_Program	Rung1	-- --